can have multiple values separated by a comma. And while a preflight request will always have an Access-Control-Request-Method header, the Access-Control-Request-Headers header is optional, and is only present if the client adds headers to the request.

ALLOWING CUSTOM HEADERS ON THE REQUEST

The server responds to the preflight request by adding the Access-Control-Allow-Headers header. The Access-Control-Allow-Headers header contains a list of headers that are allowed in requests. The following response header indicates that the client has permission to include the Timezone-Offset and the Sample-Source headers on requests:

```
Access-Control-Allow-Headers: Timezone-Offset, Sample-Source
```

If all the values in the Access-Control-Request-Headers request header match the values in the Access-Control-Allow-Headers response header, the browser is granted permission to make the request. If the browser requested a header, and that header isn't present in the Access-Control-Allow-Headers header, the request is rejected. Table 4.4 shows an example of a valid and an invalid Access-Control-Allow-Headers header.

Table 4.4 Responding to Access-Control-Request-Headers by using Access-Control-Allow-Headers. All requested headers must also be in the response for the CORS request to succeed.

Response header	Preflight request status
Access-Control-Allow-Headers: Timezone-Offset, Sample-Source	Accepted. Timezone-Offset and Sample-Source were requested, and both are present in the Access-Control-Allow-Headers header.
Access-Control-Allow-Headers: Timezone-Offset, Sample-Source, Anything-Else	Accepted. Same as the previous case. Even though the Anything-Else header is not present in the request, it's okay to specify additional values in the Access-Control-Allow-Headers header.
Access-Control-Allow-Headers: Timezone-Offset	Rejected. Both Timezone-Offset and Sample-Source were requested, but only Timezone-Offset is present in the Access-Control-Allow-Headers header.

Listing 4.7 recaps how the browser translates additional headers from code into the preflight request, and how these headers flow from preflight request to preflight response to actual request. The preflight starts in the developer's JavaScript code, which adds new headers to the `XMLHttpRequest` ❶. The browser notices that there are additional headers and puts them in the Access-Control-Request-Headers header in the preflight ❷. The server responds by including those same headers in the Access-Control-Allow-Headers response ❸. Finally, the browser validates that the headers match those requested by the developer, and sends the request to the server ❹.

Listing 4.7 Flow of additional headers on CORS requests

JavaScript code:

```
var xhr = new XMLHttpRequest();                          ❶
xhr.setRequestHeader('Timezone-Offset');
xhr.setRequestHeader('Sample-Source');
```